

of Software Project Management (SE4002)

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Do not write below this line

Attempt all the questions.

Single sided, A4 sized, hand written cheat sheet is allowed.

Course Instructor(s)

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Sessional-I Exam

Total Time (Hrs): 1
 Total Marks: 50
 Total Questions: 3

CLO #1: Understand project management principles and techniques

Q1: Mark the given statement as True/False.

10 marks

No.	Statement	True/False
1.	Staff members who are officially allocated full-time to a project may effectively be part-time due to routine work responsibilities.	T ✓
2.	The "discount rate" refers to the <u>annual</u> rate used to determine the present value of future earnings. <i>interest rate?</i>	T ✓
3.	PRINCE2 is a proprietary project management standard that requires a licensing fee for public use.	F ✓
4.	According to Parkinson's Law, providing staff with an easy target or generous deadline typically results in increased productivity and early completion <i>has no calculations & processing</i>	F ✓
5.	According to Albrecht's terminology, "External Inquiry" types are user-initiated transactions that retrieve specific data and simultaneously update the <u>system's internal files</u> . <i>X</i>	F ✓
6.	When evaluating a system, the primary financial focus should be strictly on the initial acquisition price rather than the total costs over the system's entire lifetime.	F ✓

7.	In a scenario where a supplier maintains the source code for a bespoke system, the implementation of an escrow agreement serves as a legal substitute for the customer's outright acquisition of copyright.	T ✓
8.	In the context of incremental delivery and contract law, a customer who identifies a supplier's default but fails to issue an official notification may be legally deemed to have condoned the failure, potentially forfeiting their right to seek future legal remedies for that specific breach.	T ✓
9.	While a Business Case justifies the investment, the Project Charter serves as the primary legal and organizational instrument that formally grants the Project Manager the authority to utilize company resources for project activities.	T ✓
10.	A Project Plan is a static document finalized during the initiation phase that serves primarily as a historical record of the initial schedule, rather than a dynamic tool for ongoing management.	F ✓

CLO 2: Apply approaches to manage and optimize the software development processes

Q2: A software development firm is utilizing Case-Based Reasoning (CBR) to estimate the effort required for a new E-commerce website project. To determine the similarity between projects, the team evaluates three primary parameters: the number of API integrations, the number of database tables, and a UI complexity score (measured on a scale of 1–10). The current Target Case is specified to require 4 API integrations, 12 database tables, and has a UI complexity score of 8. Within the firm's historical database, two potential matches are identified for comparison: Project B, which consisted of 3 APIs, 10 tables, and a UI complexity score of 6; and Project C, which involved 5 APIs, 15 tables, and a UI complexity score of 8. Identify which project qualifies as the "Nearest Neighbor" and provide a brief justification for why that project serves as the better source for the final estimate. Show all mathematical steps for full credit.

Project B

Project C

10 marks

no. of API ^{original} = 4
" " Database = 12
" " UI score = 8.

3 API
10 tables
6 UI

5 API
15 tables
8 UI

distance (0, B) = $\sqrt{(4-3)^2 + (12-10)^2 + (8-6)^2} = 3 \checkmark$

distance (0, C) = $\sqrt{(4-5)^2 + (12-15)^2 + (8-8)^2} = \sqrt{10} \approx 3.16227766$

since similarity of current target case and project B is greater, Project B is the nearest neighbour. (smaller distance with target case)
It may be better because it has similar complexity as target case due to which project B's effort will align with target case. ~~It has target and project C has same UI complexity and minor differences in other parameters. The distance is a good estimate because it's complexity is nearest to target.~~
It's a good estimate because it's complexity is nearest to target.

CLO 2: Apply approaches to manage and optimize the software development processes

Q3: From the given data calculate the ROI and NPV using a 10% discount rate. Decide which project is worthwhile based on calculated results. Remember to show your working. 30 marks

[5+5+5+5+5+5]

Year	Project A (USD)	Project B (USD)	Project C (USD)
0	-220,000	-180,000	-300,000
1	75,000	40,000	120,000
2	60,000	70,000	80,000
3	95,000	30,000	140,000
4	80,000	100,000	100,000
5	110,000	60,000	160,000
Net profit	200,000 <i>add all rows.</i>	120,000	300,000
ROI for project A <i>avg annual profit = 200000/5 = 40000</i> $ROI = \frac{40000}{220000} \times 100 = 18.18\%$ ROI for project B <i>avg = 120000/5 = 24000</i> $ROI = \frac{24000}{180000} \times 100 = 13.33\%$ ROI for project C <i>avg = 300000/5 = 60000</i> $ROI = \frac{60000}{300000} \times 100 = 20\%$			

Use the following for discount rate ranges

15

TABLE 2.2 NPV discount factors

Year	Discount rate (%)					
	5	6	8	10	12	15
1	0.9524	0.9434	0.9259	0.9091	0.8929	0.8696
2	0.9070	0.8900	0.8573	0.8264	0.7972	0.7561
3	0.8638	0.8396	0.7938	0.7513	0.7118	0.6575
4	0.8227	0.7921	0.7350	0.6830	0.6355	0.5718
5	0.7835	0.7473	0.6806	0.6209	0.5674	0.4972

Show working for NPV in provided tables:

Year	Project A cash flow (USD)	Discount factor @	Discounted cash flow (USD)
0	-220,000	1	-220,000
1	75,000	0.9091	68182.5
2	60,000	0.8264	49584
3	95,000	0.7513	71373.5
4	80,000	0.6830	54640
5	110,000	0.6209	68299
		NPV =	92079

-220000
x1

0.9091
x 75000

Year	Project B cash flow (USD)	Discount factor @	Discounted cash flow (USD)
0	-180,000	1	-180,000
1	40,000	0.9091	36364
2	70,000	0.8264	57848
3	30,000	0.7513	22539
4	100,000	0.6830	68300
5	60,000	0.6209	37254
		NPV =	42305

Year	Project C cash flow (USD)	Discount factor @	Discounted cash flow (USD)
0	-300,000	1	-300,000
1	120,000	0.9091	109092
2	80,000	0.8264	66112
3	140,000	0.7513	105182
4	100,000	0.6830	68300
5	160,000	0.6209	99344
		NPV	148030

$$NPV = \sum (\text{CashFlow} \times \text{discount rate})$$

→ sum of column.

Project 3 is worth while. because it has highest NPV & ROI.